

# Handoff: what was done September 15 to 19, 2026

Self-contained summary for restarting the discussion with ChatGPT

September 19, 2026

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## 0. How to read this

This is a complete record of five days of work on the housing and fertility model, written so that someone who did not follow the sessions can pick up the discussion. Every number is a steady-state result at the paper’s retained parameters unless it says otherwise. Nothing below changed the paper’s code defaults, targets, or calibration; everything ran as switches or overrides in a sandbox. Paths are relative to the repository root / Users/tommasodesanto/Desktop/Projects/Fertility/Fertility\_Spring26.

The model in one paragraph. Households live in four-year periods from 18 to death, choose consumption, liquid savings, rent-or-own and housing size in rooms, and whether to try for a child. Children ever born is  $n$ ; children currently at home is  $m$ . Utility is CRRA over a Cobb–Douglas composite of consumption and

housing services in excess of a family space floor  $\bar{h}(m)$ , deflated by an equivalence scale, plus a linear child benefit  $\psi m$  and a first-birth cost  $\xi$ . Owners pay a down payment  $(1 - \phi)Ph$  at purchase with  $\phi = 0.80$  fixed, can borrow against the house up to  $\phi Ph$  every period, and have no unsecured credit ( $\lambda_d = 0$ ). Rentals exist only up to six rooms; owner sizes are  $\{2, 4, 6, 8, 9.5, 11\}$ . Housing supply is  $H^s = H_0 P^\eta$ , rent is a user cost, pensions are pay-as-you-go, property tax is rebated equally. Children leave home at a constant rate  $\mu = 4/18$  per period regardless of their age or the parent's age. Earnings are a permanent type times a persistent AR(1). The paper's mechanism, as stated on the September 14 slides, is that the down payment constrains young households' access to family-sized housing and so delays and reduces births.

## 1. Timeline

**September 15.** A referee-style structural review of the model (labels P1–P5 preferences, F1–F5 fertility and family, H1–H6 housing and finance, D1–D4 demographics, E1–E2 earnings, X1–X3 cross-cutting), each claim checked against the code with file and line. ChatGPT Pro produced a parallel review from the same prompt. The two were merged into one document with agreements condensed, disagreements shown with both positions, and one-sided points attributed. That document has a follow-up log that has been the single tracking file since; every study below has a dated row in it.

- docs/model/structural\_model\_review\_consolidated\_20260915.md (PDF in output/pdf/). Sections per label, a “Where the mechanism stands” section, the follow-up log, a combined table.
- The two source reviews: structural\_model\_review\_fable\_20260915.md, structural\_model\_review\_chatgpt\_20260915.md, same folder.

**September 16.** Cluster down. Reading on P1, the child term in utility: how the literature models the benefit and cost of children, three candidate specifications (docs/model/child\_cost\_utility\_lit\_review\_20260916.md). Two questions from Guido added to the issue ledger (transaction volumes; rooms per person over time; docs/model/POST\_PRESENTATION\_ISSUES.md M40, M41). Decision to build a test environment: change one assumption, re-solve the general equilibrium steady state, get moments against targets and policy plots, four output files, runnable by the author.

**September 17.** The sandbox was built and, after a day lost to a baseline that did not reproduce the slides, made reliable. Provenance settled: the paper's code is the tag paper-baseline-2026-09-14, main was reconciled to it, and the September 14 initial state replays exactly on the cluster (loss 179.298, all 13 moments and 17 parameters). A byte copy of the presented code is under calibration\_archive/legacy\_presentation\_20260914/. Standing rule adopted: no shock re-estimation, no transition, no policy run until the economics decisions are made; the recovery calibration batch stays stopped. Evening and night: five model switches built as default-off options, the frictionless benchmark, the who-is-constrained diagnostic, the scarce-space test, the maturation study, and single-switch steady states at fixed and re-normalized child preference.

**September 18.** Situation report at 10:00 (docs/model/situation\_report\_20260918.md, PDF in output/pdf/). Two long cluster runs resubmitted after a queue cancellation. Two literature and evidence memos by a strong subagent, checked against the sources by me: how many young households are down-payment constrained, and whether tenure segmentation by size is real. A cap-at-eight-rooms test run.

**September 19.** Cluster runs collected, cap test collected, this handoff.

## 2. The sandbox and what “baseline” means

code/model/sandbox/ (README there). `run_ss.py --spec <name>` reads a YAML spec of overrides, assembles the paper’s parameters with the paper’s own recipe functions, solves the stationary general equilibrium on the full production grid, and writes `summary.md`, `moments.csv`, `parameters.csv`, `graphs.pdf` to `output/model/sandbox/<spec>/`. About three minutes per solve on the laptop. `psi_mode: fixed` keeps the child benefit level  $\psi$  at its retained value (0.149); `psi_mode: root` re-sets  $\psi$  so completed fertility equals 2.1 (eight to eleven solves). Specs live in `code/model/sandbox/specs/`, in fixed and root twins.

Caveat that applies to every number below. The sandbox baseline is close to but not equal to the slides’ state (ownership 0.46 against 0.54 on the slides; the launch recipe starts from a checkpoint that exists only on the cluster). Rule: the cluster replay is the exact tool for levels; the sandbox is the tool for differences. Losses are on the sandbox’s twelve-row objective and are comparable across sandbox rows only, never with the slides’ 179.

Sandbox baseline,  $\psi$  root: loss 2607,  $\psi$  0.190, childless 0.201 (target 0.198), mean first-birth age 26.1 (26.0), first births at 30+ 0.234 (0.249), wealth to earnings 5.14 (6.15), old-age p90/p50 4.32 (3.52), mean rooms 5.75 (5.56), ownership 30–55 0.455 (0.648), first-birth rooms response 1.12 (0.72), rooms gap three-plus versus one-to-two children 0.24 (0.35), recent-parent ownership gap 0.46 (0.16).

## 3. The mechanism test (the result that hurts)

**Frictionless benchmark, fixed  $\psi$ .**

| Moment                        | Baseline | No down payment<br>( $\phi = 1$ ) | Unsecured credit line<br>(five years’ earnings) |
|-------------------------------|----------|-----------------------------------|-------------------------------------------------|
| Completed fertility           | 1.872    | 1.866                             | 1.948                                           |
| Childless 40–44               | 0.239    | 0.241                             | 0.225                                           |
| Mean first-birth<br>age       | 26.96    | 26.97                             | 25.79                                           |
| Ownership 30–55               | 0.459    | 0.557                             | 0.450                                           |
| Wealth / earnings             | 5.18     | 5.07                              | 4.43                                            |
| First-birth rooms<br>response | 1.07     | 1.15                              | 1.17                                            |

Removing the down payment raises ownership by ten points and does nothing to births or their timing. Giving households unsecured credit raises fertility and brings first births forward by more than a year and does nothing to ownership.

**Who is constrained.** Share of households aged 26 to 38 with zero or one child at home whose modal tenure-and-size choice changes when the down payment is removed, at the same state: 4.5 percent (a stock object over the baseline distribution). Lowering the supply scale until mean rooms equal the data (5.56, price up 4 percent) drops that to 2.8 percent and leaves fertility flat with and without the down payment (1.840 against 1.835). The author’s prior that the rooms overshoot hides the mechanism was rejected on this one-parameter test; the refit version, where the space floor, child cost and wedge move together, is untested.

**Why.** A renter can have six rooms and the parents’ space floor is 2.3 rooms, so one- and two-child families fit in a rental; owner-only sizes start at eight rooms and matter for three-plus families. What blocks an early

birth is that a renter cannot borrow at all and must save the child's cost in advance. That is a precautionary channel, real, but not the collateral story on the slides.

**A second structural fact.** Children at home are net costs in the value function at every state:  $V$  falls when a child stays one more period. Fertility is held up by the taste shocks and by  $\psi$  being set to hit 2.1. Any change that lengthens dependency or adds a child cost must be compared with  $\psi$  re-normalized, never at fixed  $\psi$  (a fixed- $\psi$  run of the maturation law collapsed fertility to 0.49 for this reason).

**What survives.** Easier mortgages raise ownership, not births. Easier unsecured credit raises births, not ownership. Housing costs lower fertility through the price of space. The slide's mechanism sentence has to change. The model, the estimation machinery, the transition solver and the property-tax experiment all stand.

## 4. Switches built (all default off, package tests 227 passed)

Each is a parameter in `code/model/intergen_eqscale_seq_optimized/` that, when at its default, leaves every array bitwise identical (tested). Built by a delegated coder from exact specifications, verified line by line by me.

1. **Parent-age maturation** (F3). Children leave home at rate 0.05 per period while the parent is under 34, rising linearly to certain exit at parent age 62; newborns exempt in their birth period. Replaces the memoryless constant rate, keeps the state space. Required an exact re-solve of the second housing stage in the fertile ages (a shortcut proposed by the coder was rejected).
2. **Child earnings penalty** (P1, S3). Working-age after-tax earnings times  $1 - \tau_c(m)$ , spec  $[0, 0.2, 0.2, 0.2]$ ; pensions and the payroll base unpenalized.
3. **Mortgage block** (H1). Origination-only collateral test (a stayer cannot increase debt; free cash-out removed) and amortization of 11 percent of principal per period.
4. **Size-dependent rental wedge** (H3/H6). Rent per room  $r + w_0 + w_1 \max\{0, h - 6\}$ , spec  $w_0 = 0.02, w_1 = 0.05$ , with the cap removed (hR\_max 11) and the owner premium set to one.
5. **Estate receiver** (P5). Estates valued net of selling cost and paid as an equal transfer to households aged 45 to 65, an outer fixed point.

Parameter-only specs: modest unsecured credit line ( $\lambda_d = 0.1875$ , three quarters of a quarter of mean earnings, Kaplan–Violante), E6b earnings (one PSID decomposition, both components after-tax), Boar–Gorea–Midrigan earnings (no permanent types, five states), tenure shock  $\kappa_H = 0$ , concave (log) child benefit, cap at eight rooms.

## 5. Results table, $\psi$ re-normalized to 2.1

Targets: childless 0.198, first-birth age 26.0, ownership 0.648, old p90/p50 3.52, rooms response 0.72, recent-parent gap 0.16. Baseline loss 2607.

| Change   | Loss | $\psi$ | Childless | First-birth age | Ownership 30–55 | Old p90/p50 | Rooms resp. | Recent-parent gap |
|----------|------|--------|-----------|-----------------|-----------------|-------------|-------------|-------------------|
| Baseline | 2607 | 0.190  | 0.201     | 26.1            | 0.455           | 4.32        | 1.12        | 0.47              |

| Change                                                       | Loss | $\psi$ | Childless | First-<br>birth age | Ownership<br>30–55 | Old p90/<br>p50 | Rooms<br>resp. | Recent-<br>parent<br>gap |
|--------------------------------------------------------------|------|--------|-----------|---------------------|--------------------|-----------------|----------------|--------------------------|
| Rental<br>wedge,<br>no cap,<br>$\chi = 1$                    | 402  | 0.205  | 0.208     | 26.3                | 0.730              | 3.83            | 0.97           | 0.28                     |
| Rental<br>cap at<br>8<br>rooms<br>only                       | 489  | 0.189  | 0.202     | 26.1                | 0.300              | 4.38            | 1.36           | 0.23                     |
| All<br>switches<br>to-<br>gether                             | 962  | 0.480  | 0.223     | 27.6                | 0.426              | 4.44            | 0.97           | 0.27                     |
| Earnings<br>penalty<br>20 %                                  | 1746 | 0.315  | 0.228     | 27.1                | 0.416              | 4.46            | 0.85           | 0.39                     |
| Mortgage<br>block                                            | 2050 | 0.189  | 0.201     | 26.1                | 0.349              | 5.05            | 1.08           | 0.42                     |
| Boar-<br>Gorea-<br>Midrigan<br>earn-<br>ings,<br>no<br>types | 2100 | 0.156  | 0.144     | 27.0                | 0.499              | 2.92            | 1.24           | 0.41                     |
| Concave<br>child<br>benefit<br>(log)                         | 2111 | 0.199  | 0.210     | 25.9                | 0.449              | 4.15            | 1.10           | 0.43                     |
| E6b<br>earn-<br>ings,<br>one de-<br>compo-<br>sition         | 2325 | 0.180  | 0.184     | 26.6                | 0.477              | 3.38            | 1.14           | 0.44                     |
| Tenure<br>shock<br>$\kappa_H =$<br>0                         | 2379 | 0.190  | 0.202     | 26.1                | 0.451              | 4.30            | 1.12           | 0.45                     |
| Parent-<br>age<br>matu-<br>ration                            | 2464 | 0.167  | 0.210     | 26.5                | 0.449              | 4.36            | 1.39           | 0.45                     |

| Change                                      | Loss | $\psi$ | Childless | First-<br>birth age | Ownership<br>30–55 | Old p90/<br>p50 | Rooms<br>resp. | Recent-<br>parent<br>gap |
|---------------------------------------------|------|--------|-----------|---------------------|--------------------|-----------------|----------------|--------------------------|
| Modest<br>unse-<br>cured<br>credit<br>line  | 2539 | 0.187  | 0.202     | 25.8                | 0.446              | 4.32            | 1.15           | 0.46                     |
| Estate<br>trans-<br>fer to<br>ages<br>45–65 | 2582 | 0.181  | 0.193     | 26.4                | 0.446              | 4.07            | 1.06           | 0.46                     |

Readings, kept short.

- The wedge is the single largest improvement, and it makes ownership come from the rent-to-price ratio rather than an owner taste premium. Its intercept is too strong at 0.02 (ownership 0.73) and would be calibrated.
- The cap at eight rooms, on the paper’s model, moves ownership (0.46 to 0.31) and not fertility (1.876 against 1.872 at fixed  $\psi$ ; childless and first-birth age unchanged). A July 1 internal audit on an earlier package had found the cap to be a strong fertility lever; that does not carry over.
- The switches do not add up. All together give 962 with  $\psi$  at 0.48, two and a half times the baseline root, first-birth age 27.6 against 26.0 and rooms 5.25 below target. The penalty and the maturation law each need a higher  $\psi$  and together overshoot the timing rows.
- The earnings penalty alone halves the loss, moves first births later and the rooms response toward target, and needs  $\psi$  to double.
- The concave benefit puts first-birth age exactly on target with fewer large families.
- Permanent earnings types are what the old-age wealth tail buys; without them the tail is 2.9 against 3.5 and childlessness falls to 0.14.
- The mortgage block lowers ownership by ten points and does nothing to fertility; it makes the finance side standard and the Coven comparison like for like.
- The tenure shock is inert; the estate receiver is a small change.

## 6. The maturation study (F3)

The constant exit rate gives 40 percent too few children at home at parent ages 30 to 42 and keeps adult “dependents” past 58 (ACS comparison in `output/model/sandbox/dependents_by_parent_age/`). The parent-age law with the newborn exemption matches the ACS profile at 22 to 34 and is zero after 62, with  $\psi$  re-normalized (`output/model/sandbox/maturation_root/`). A theory note on the trade-off between the two laws is `docs/model/f3_maturation_tradeoff_note_20260917.md`. Recommended: adopt the parent-age law. Not decided.

## 7. The two evidence memos (September 18)

Both written by a strong subagent, both with a lead review note at the top listing which sources were opened and which claims I verified myself. Both are information, not conclusions; the author wants to discuss each with data and without rush.

**How many are constrained** (docs/model/evidence\_constrained\_share\_20260918.md). Four objects get called “the constrained share”: who cannot meet 20 percent today (about two thirds of young renters and recent movers, an accounting object); who cannot meet the 3.5 percent FHA minimum (much smaller, no model counterpart); whose choice actually changes when the constraint moves (the marginal group, the only one comparable to the model’s 4.5 percent); which buyers got family help (about 20 percent of first-time buyers). Haurin, Hendershott and Wachter (NLSY, ages 20 to 33, verified in the text): 37 percent constrained, ownership probability 0.20 to 0.10 and 0.52 to 0.34 when constrained, so a marginal share of about 4 to 7 percent. Kaplan, Mitman and Violante (verified): “very few households are constrained in this way: rather than buying excessively small houses, they prefer to rent a house of the desired size”; their credit relaxation raises ownership by about 3 percent. The model’s 4.5 percent sits in that range. Proposed disciplining moment: the SCF share of renters aged 26 to 38 whose liquid assets fall below 20 percent of the local median price, a level object computable in model and data; hold the marginal share out as validation; Fuster and Zafar’s renter willingness-to-pay response to a lower down payment as a behavioral check (resolve at  $\phi = 0.95$ ).

**Tenure segmentation** (docs/model/evidence\_tenure\_segmentation\_20260918.md). Renter share by bedrooms (ACS 2024): about 86, 85, 55, 20, 11, 8 percent from studio to five bedrooms. Single-family homes are 14 percent renter-occupied. Roughly three million single-family units converted to rental use in 2007 to 2011, so the boundary is not technological. The only direct evidence that large rentals are a different good is Halket, Nesheim and Oswald on London: owner share rises from 33 percent below 50 square meters to 90 percent above 100, and unobserved rental quality falls with size. Our own within-cell estimate of the rent of an extra family-sized room is zero, so scarcity is availability, not price. Kaplan, Mitman and Violante use the same partial segmentation (top size classes owner-only). Greaney, Parkhomenko and Van Nieuwerburgh set the rental maximum equal to the owner maximum (verified in the local text), so an internal July note attributing our six-room cap to a “DUE rule” is wrong. The model’s hard zero above six rooms is false against about 6 percent of prime-age childless renters at seven or more rooms. Honest framing: the cap is a calibrated availability friction; the wedge is the version without a hard zero, and it is the switch that fits best.

## 8. Decisions pending (author), with the recommendation on file

1. **P1, the child term.** To be rethought. Facts: children at home are net costs at every state; the intensive margin is carried by the taste scales; the concave benefit fixes timing; the earnings penalty fixes the income gradient at the price of a doubled  $\psi$ . Candidates in the P1 reading: keep linear and say so; concave benefit; earnings penalty; the Sommer–Sullivan–Kindermann style weighting was analyzed and makes child costs rise with  $m$ .
2. **F3.** Adopt the parent-age law (recommended).
3. **H3/H6.** Wedge in place of cap plus owner premium, calibrated on the AHS renter share by size (recommended); to be discussed with data.
4. **H1.** Mortgage block (recommended); a modest unsecured credit line (recommended, cheapest timing improvement).
5. **E1.** Keep permanent types with the E6b decomposition (recommended).
6. **F2.** Tenure shock to zero (recommended).

7. **P5.** Net-of-selling-cost valuation regardless; the receiver is a welfare-accounting choice.
8. **The constrained-share question and tenure segmentation:** to be discussed in turn, with data, before anything is decided.

After the decisions: one refit on the cluster under the same target contract, then the frictionless and constrained diagnostics again on the refit, then the mechanism paragraph rewritten around what the model does, then the history refit and the policy transition.

## 9. Rules that were set during the week

- Do not change the paper’s code defaults; every change is a default-off switch or a sandbox override.  $\phi = 0.80$  is fixed.
- No shock re-estimation, transition, or policy run until the economics decisions are made. The recovery calibration batch (job 17858740) is stopped and stays stopped.
- Real runs go to the cluster; the laptop is for minutes-scale sandbox solves and analysis. Two of the  $\psi$ -root runs (estate, all switches) took 2.2 and 8.8 hours on the cluster.
- Coding is delegated with exact specifications and reviewed line by line; model-critical numerics are verified against the math before they are trusted.
- Never use the word “parity” in prose. Plain English, no em dashes.
- One tracking document: the consolidated review’s follow-up log.

## 10. Open items and known problems

- Combined  $\phi = 1$  plus credit line trips a dead-mass invariant in the solver; unsolved, not needed for any conclusion.
- The sandbox baseline differs from the slides in two wealth-distribution rows because of the cluster-only checkpoint; the regression gate against the deck’s levels stays open.
- A stray git ref (refs/remotes/origin/main 2) blocks automatic repacking; harmless, left for the Codex session.
- The July 2026 internal cap note and the audit numbers on the earlier package should not be cited for the paper’s model.
- Two Kaplan, Mitman and Violante claims in the tenure memo (their Table 6 renter shares; the segmentation robustness) are attributed to the published version and were not verified by me.
- The proposed SCF liquid-assets moment does not exist as a published statistic; it would be built from microdata.

## 11. File map

- Reviews: docs/model/structural\_model\_review\_consolidated\_20260915.md and the two source reviews; PDFs in output/pdf/.
- Situation report: docs/model/situation\_report\_20260918.md.
- P1 reading: docs/model/child\_cost\_utility\_lit\_review\_20260916.md.
- F3 note: docs/model/f3\_maturation\_tradeoff\_note\_20260917.md.
- Evidence memos: docs/model/evidence\_constrained\_share\_20260918.md, docs/model/evidence\_tenure\_segmentation\_20260918.md.
- Sandbox: code/model/sandbox/ (README, run\_ss.py, specs/, diagnostics\_constrained.py, diagnostics\_dependents\_by\_age.py).

- Results: output/model/sandbox/ (frictionless/, constrained/, switches\_fixed/, switches\_root/psi\_root\_comparison.md, maturation\_root/, torch\_psi\_root/, cap\_eight\_psi\_\*).
- Switch specifications given to the coder: docs/prompts/TASK\_muse\_\*.md.
- Provenance: tag paper-baseline-2026-09-14; output/model/paper\_baseline\_sep14/replay\_20260917/README.md; calibration\_archive/legacy\_presentation\_20260914/.